

this important area of nursing.

The references¹ that have been included illustrate methods and research studies which have been utilized by research investigators. It is not the purpose of this column to indicate areas of nursing needing research, nor to submit these sources as necessarily representing the best techniques or methods which may be employed in any one research study. Neither are these sources submitted with the purpose of presenting conclusions from any specific research investigation. Rather they are presented as *starting points for the investigator who is planning to undertake research in the area of nursing service*. It is expected that the reader will study these sources critically, evaluate them, and utilize them only as they fit in with his or her purposes.

Although nursing service administration is a field in which there is currently considerable interest and research activity, it is also one in which much more research is still needed. Readers are therefore asked to send to Researcher's Bookshelf any source material in this area which they have found helpful and which is not included in the list below.

AMERICAN MANAGEMENT ASSOCIATION. *How to Establish and Maintain a Personnel Department.* (Research Report Number Four) 3rd edition. New York: American Management Association, 1946. 116 pages.

Within the contents of this booklet, personnel research projects are suggested. These projects include: (1) the efficacy of individual incentives on productivity; (2) the causes of turnover, and resultant adjustments in personnel policies; (3) the necessity for proper measures for the selection of supervisory personnel; and (4) the evaluation of personnel attitude as a means of developing good morals.

ANDERSON, E. H. "The Meaning of Scientific Management." *Harvard Business Review* 27:678-692, November 1949.

¹ Contributions to this list of references were made by a number of individuals engaged in research in nursing service administration. Special acknowledgment for assistance in compiling this bibliography is due to: Charlotte Seyfer, associate professor and director of nursing service administration, Catholic University of America, Washington, D. C.; Margaret Mary Moleaky, secretary, Division of Nursing Service, Catholic Hospital Association; Sister Agnita Claire, director of nursing service and personnel director, St. Mary's Hospital, Kansas City, Mo.; and Louise M. Schmitt, director of nursing service administration, State University of Iowa, Graduate Division, College of Nursing.

This article presents several branches of scientific management and discusses the significant developments in each. Some of the phases included are: the economic, the administrative, the engineering, the psychological, and the sociological. Many of the developments in the field of scientific management may be significantly applied to the management of nursing service. Incentive techniques, the value of proper physical facilities, and studies of fatigue are included among the interesting topics which can be very practically applied to hospital situations.

ANGOFF, WILLIAM H. "Reducing Accidents by Collecting Data." *Personnel Journal* 27:387-391, March 1949.

This article, written by a research psychologist of the Human Engineering Branch at the Office of Naval Research in Port Washington, N. Y., sets forth a plan by which accidents can be reduced if scientific methods of measurement are used. Facts, specific knowledge, and plans for action are needed before any industrial plant can have a high safety rating.

ANTHONY, ROBERT N. and DAY, JOHN S. *Management Controls in Industrial Research Organizations.* Cambridge, Massachusetts: Harvard University, Division of Research, 1952. 537 pages.

This study presents the problems of administrative control of scientific research activities in American industry. The need for control is necessarily intensified because of the nonstandard and individualistic nature of scientific research. Information contained in this book will be helpful to the nurse researcher as well as to the hospital or agency in which a research project is being conducted.

BARNES, RALPH M. *Work Measurement Manual.* 4th edition, Dubuque, Iowa: Wm. C. Brown Co., 1951. 297 pages.

This volume explains briefly the importance of time study, the procedure commonly used in making such a study, the work measurement investigations now being conducted by Professor Barnes, and some of the preliminary findings. It tells how to conduct a community time study survey and gives the results of several such surveys. Five thousand people from more than 350 different industrial concerns in the United States and Canada have participated in the Barnes Work Measurement Project. The manual describes in detail how an organization may proceed to check the ability of its time study men, to set time standards, and to improve the accuracy and consistency of such standards. It also explains how these standards may be compared with the "national standards" being developed by Professor Barnes.

BARNES, RALPH M. and ENGLERT, N. A. 5th edition. *Bibliography of Industrial Engineering and Management Literature to January 1, 1946.* Dubuque, Iowa: Wm. C. Brown Co., 1946. 136 pages.

More than 1200 important books and bulletins in the general field of industrial engineering and management are listed in this volume. The bibliography also contains reference to over 3000 articles and papers on motion and time studies and related subjects, and includes a classified index of articles and papers under 47 main headings and 79 subheadings. The bibliography is not annotated; it contains only the name of the author, name of the article, publisher, volume number, and date. This is the only comprehensive guide to recent periodical literature in the field of motion and time study.

BARNES, RALPH M. *Work Methods Training Manual.* Dubuque, Iowa: Wm. C. Brown Co., 1950. 320 pages.

A methods improvement training program, to be most effective, should be developed to meet the needs of the particular group to receive the training. Two program outlines are included in this volume to show in detail the type of course that has proved effective in the past. It is not expected that these outlines will be used exactly as they are presented but rather it is hoped that they may serve as a guide to others who are confronted with the problem of designing a program for their own particular group.

BRAVFIELD, ARTHUR H. and ROTHE, HAROLD F. "An Index of Job Satisfaction." *Journal of Applied Psychology* 35:333-337, October 1951.

Job satisfaction is often based on subjective judgment. In this article, the author presents a procedure in which 18 items are listed on a "job satisfaction blank," which proved to be an attitude scale of high validity and reliability.

BRISLEY, C. L. "Work Sampling." *Factory Management and Maintenance* 110:83-89, July 1952.

This article presents a new method of sampling called the "work sampling" or "ratio delay method," which has been used in industry to help cut costs and boost production. This method was used in a recent study at the State University of Iowa hospitals to find out how nursing employees spend their time.

ECKLEY, G. O. "Setting a Pattern for Procedures Analysis." *Controller* 17:425-427, September 1949.

A check-list is used as a research tool in improving reports, policies, productivity, scheduling, and methods. The author contends that a procedures problem necessitates correct reasoning for a worthwhile solution. To be avoided are generalizations, opinions, and casual observations.